

Constructive Method for Goldbach Representations Yielding Generated-Family Bounds and a Prime-Selector Heuristic

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Abstract

Goldbach's binary problem asks whether every even integer greater than two is a sum of two primes. This paper studies a direct two-stage interval construction that, from a sufficiently large even input $2n$, selects a prime p' just below $2n-5$, forms the residual $M=2n-p'$, and then selects a prime q' above M . Writing $q'=M+2c$ yields a Goldbach representation $p'+q'=2n+2c$. The earlier shift identity $p'=p+2k$, $q'=q-2k+2c$ is retained only as an equivalent parametrization when an input representation $p+q=2n$ is available. Using the Baker-Harman-Pintz exponent $\theta=0.525$, together with a finite treatment below the asymptotic threshold, the iterated construction generates $\Omega(x^{0.724375})$ Goldbach even integers up to x . Under an explicit unproved Cramér-type $O((\log x)^2)$ gap assumption, the two-scale argument gives $c=O((\log \log n)^2)$ and $\Omega(x/(\log \log x)^2)$ generated values. A separate selector heuristic uses the Hardy-Littlewood singular series but does not assume that candidate events are independent as a theorem. The revised numerical experiment reports the complete distribution of $S_G(n)$ and the minimal shift c for 900,001 inputs. Neither full Goldbach nor generated density one is proved.

Keywords: Goldbach conjecture; primes in short intervals; prime gaps; Cramér model; singular series; constructive number theory; probabilistic heuristic.

Introduction and Literature Review

Exceptional sets in the binary Goldbach problem

Hardy and Littlewood supplied the classical circle-method heuristic for binary Goldbach, including the arithmetic correction now expressed through the singular series ((G. H. Hardy, J. E. Littlewood. Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes. *Acta Mathematica*. Vol. 44, pp. 1-70, 1923. <https://doi.org/10.1007/BF02403921>)). On the unconditional side, Montgomery and Vaughan proved a power-saving estimate for the exceptional set: if $E(x)$ counts even integers up to x that are not sums of two primes, then $E(x) \ll x^{1-\delta}$ for some $\delta > 0$ ((H. L. Montgomery, R. C. Vaughan. The exceptional set in Goldbach's problem. *Acta Arithmetica*. Vol. 27, pp. 353-370, 1975)). Li obtained a later refinement of the exceptional-set estimate ((H. Li. The exceptional set of Goldbach numbers (II). *Acta Arithmetica*. Vol. 92(1), pp. 71-88, 2000. <https://doi.org/10.4064/aa-92-1-71-88>)). Matomäki studied the exceptional set in short intervals ((K. Matomäki. On the exceptional set in Goldbach's problem in short intervals. *Monatshefte für Mathematik*. Vol. 155(2), pp. 167-189, 2008. <https://doi.org/10.1007/s00605-008-0561-1>)), Lu obtained a further exceptional-set bound ((W. C. Lu. Exceptional set of Goldbach number. *Journal of Number Theory*. Vol. 130(10), pp. 2359-2392, 2010. <https://doi.org/10.1016/j.jnt.2010.03.017>)), and Grimmelt studied Goldbach numbers in short intervals ((L. Grimmelt. Goldbach numbers in short intervals. *Annali della Scuola Normale Superiore di Pisa - Classe di Scienze*. Vol. 23(3), pp. 1395-1416, 2022. https://doi.org/10.2422/2036-2145.202002_001)). A 2026 paper of Bhowmik and Grimmelt reviews this line and adds an explicit major-arc formula and a conditional observation concerning exceptional zeros ((G. Bhowmik, L. Grimmelt. The exceptional set of the Goldbach problem. *Analysis Mathematica*, 2026. <https://doi.org/10.1007/s10476-026-00191-z>)).

What is known from this literature is therefore much stronger than mere infinitude: Goldbach-representable even integers have density one. What remains open is the elimination of the exceptional

set altogether. The present paper does not improve $E(x)$. Its unconditional result concerns a particular iteratively generated subfamily, so its counting theorem must be compared with exceptional-set theory only to clarify scope, not as a competing density-one result.

Primes and Goldbach numbers in short intervals

The unconditional interval input used in this paper is Baker-Harman-Pintz: for all sufficiently large x , the interval $[x-x^{0.525}, x]$ contains a prime ((R. C. Baker, G. Harman, J. Pintz. The difference between consecutive primes, II. Proceedings of the London Mathematical Society. Vol. 83(3), pp. 532-562, 2001. <https://doi.org/10.1112/plms/83.3.532>)). Explicit interval estimates have continued to be developed; for example, Cully-Hugill and Lee give explicit prime-location estimates ((M. Cully-Hugill, E. S. Lee. Explicit interval estimates for prime numbers. Mathematics of Computation. Vol. 91(336), pp. 1955-1970, 2022. <https://doi.org/10.1090/MCOM/3719>)). Recent work has also obtained conditional short-interval results directly for Goldbach numbers. Assuming the Riemann Hypothesis, Cully-Hugill and Dudek proved that every interval $(x, x+9696(\log x)^2]$ contains a Goldbach number ((M. Cully-Hugill, A. W. Dudek. An explicit mean-value estimate for the prime number theorem in intervals. Journal of the Australian Mathematical Society. Vol. 117(1), pp. 1-15, 2024. <https://doi.org/10.1017/S1446788723000113>)), and Chirre and Hagen improved the constant 9696 to 123 ((A. Chirre, M. V. Hagen. On Goldbach numbers in short intervals. Research in Number Theory. Vol. 12, Article 82, 2026. <https://doi.org/10.1007/s40993-026-00767-6>)). In a different direction, Guth and Maynard proved new zero-density estimates that imply asymptotics for primes in intervals of length $x^{(17/30+o(1))}$ ((L. Guth, J. Maynard. New large value estimates for Dirichlet polynomials. Annals of Mathematics. Vol. 203(2), pp. 623-675, 2026. <https://doi.org/10.4007/annals.2026.203.2.6>)).

These statements answer different questions. Baker-Harman-Pintz gives the one-sided existence theorem used here; Guth-Maynard concerns asymptotic prime counts on a longer power scale; the 2024 and 2026 Goldbach-interval papers are conditional on RH and concern the presence of at least one Goldbach number in a nearby interval. The construction below instead applies a prime-in-interval input twice, at two different scales, to control the jump it generates.

Prime gaps and local structure

Small-gap results show that bounded gaps occur infinitely often, but they do not furnish a uniform bound for the next prime after every integer. The Goldston-Yıldırım and Goldston-Pintz-Yıldırım correlation methods initiated the modern small-gap framework ((D. A. Goldston, C. Y. Yıldırım. Higher correlations of divisor sums related to primes III: small gaps between primes. Proceedings of the London Mathematical Society. Vol. 95(3), pp. 653-686, 2007. <https://doi.org/10.1112/plms/pdm021>)) ((D. A. Goldston, J. Pintz, C. Y. Yıldırım. Primes in tuples I. Annals of Mathematics. Vol. 170(2), pp. 819-862, 2009. <https://doi.org/10.4007/annals.2009.170.819>)) ((D. A. Goldston, J. Pintz, C. Y. Yıldırım. Primes in tuples II. Acta Mathematica. Vol. 204(1), pp. 1-47, 2010. <https://doi.org/10.1007/s11511-010-0044-9>)); Zhang, the Polymath project, and Maynard established and refined bounded-gap results ((Y. Zhang. Bounded gaps between primes. Annals of Mathematics. Vol. 179(3), pp. 1121-1174, 2014. <https://doi.org/10.4007/annals.2014.179.3.7>)) ((D. H. J. Polymath. Variants of the Selberg sieve, and bounded intervals containing many primes. Research in the Mathematical Sciences. Vol. 1, Article 12, 2014. <https://doi.org/10.1186/s40687-014-0012-7>)) ((J. Maynard. Small gaps between primes. Annals of Mathematics. Vol. 181(1), pp. 383-413, 2015. <https://doi.org/10.4007/annals.2015.181.1.7>)). Large-gap results of Maynard, Ford-Green-Konyagin-Tao, and Ford-Green-Konyagin-Maynard-Tao demonstrate that unusually long prime-free intervals also occur ((J. Maynard. Large gaps between primes. Annals of Mathematics. Vol. 183(3), pp. 915-933, 2016. <https://doi.org/10.4007/annals.2016.183.3.3>)) ((K. Ford, B. Green, S. Konyagin, T. Tao. Large gaps between consecutive prime numbers. Annals of Mathematics. Vol. 183(3), pp. 935-974, 2016.

<https://doi.org/10.4007/annals.2016.183.3.4>)). ((K. Ford, B. Green, S. Konyagin, J. Maynard, T. Tao. Long gaps between primes. *Journal of the American Mathematical Society*. Vol. 31(1), pp. 65-105, 2018. <https://doi.org/10.1090/jams/876>)). Fourier-optimization work provides another perspective on quantitative gap questions ((E. Carneiro, M. B. Milinovich, K. Soundararajan. Fourier optimization and prime gaps. *Commentarii Mathematici Helvetici*. Vol. 94(3), pp. 533-568, 2019. <https://doi.org/10.4171/CMH/467>)).

Thus a single worst-case gap function suppresses information about the range of available primes inside a search window. That distinction matters here: the deterministic part of the paper uses a worst-case interval guarantee, whereas the unit-step question is a selector problem that asks whether at least one first-stage prime has a favorable complementary residual.

Cramér-type gap heuristics

Cramér introduced a random model for primality and used it to motivate a maximal-gap scale of order $(\log x)^2$ ((H. Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*. Vol. 2(1), pp. 23-46, 1936. <https://doi.org/10.4064/aa-2-1-23-46>)). Granville explained important arithmetic limitations of the naive independence model and refined the heuristic picture ((A. Granville. Harald Cramér and the distribution of prime numbers. *Scandinavian Actuarial Journal*. Vol. 1995(1), pp. 12-28, 1995. <https://doi.org/10.1080/03461238.1995.10413946>)). Granville and Lumley later compared short-interval heuristics with computations ((A. Granville, A. Lumley. Primes in short intervals: Heuristics and calculations. *Experimental Mathematics*. Vol. 32(2), pp. 378-404, 2023. <https://doi.org/10.1080/10586458.2021.1927256>)), while Banks, Ford, and Tao introduced a sieve-aware probabilistic model that more faithfully incorporates residue-class obstructions and yields heuristic information on large gaps ((W. Banks, K. Ford, T. Tao. Large prime gaps and probabilistic models. *Inventiones Mathematicae*. Vol. 233(3), pp. 1471-1518, 2023. <https://doi.org/10.1007/s00222-023-01199-0>)).

No uniform $O((\log x)^2)$ gap theorem is known for the primes. Accordingly, every use of that scale below is labeled an assumption or heuristic. The paper's conditional result is not presented as a consequence of the prime number theorem, RH, or the Baker-Harman-Pintz theorem.

Singular series and probabilistic selector models

For binary Goldbach, Hardy-Littlewood's heuristic predicts a representation count with a multiplicative local correction depending on the prime divisors of the target even integer ((G. H. Hardy, J. E. Littlewood. Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes. *Acta Mathematica*. Vol. 44, pp. 1-70, 1923. <https://doi.org/10.1007/BF02403921>)). Modern work continues to analyze these local factors: Goldston and Suriajaya study a Cesàro mean of the prime-pair singular series ((D. A. Goldston, A. I. Suriajaya. The error term in the Cesàro mean of the prime pair singular series. *Journal of Number Theory*. Vol. 227, pp. 144-157, 2021. <https://doi.org/10.1016/j.jnt.2021.03.004>)); Leung proves conditional Poisson point-process limits under a uniform prime-k-tuple hypothesis and obtains mean-value results for singular series ((S.-K. Leung. Pseudorandomness of primes at large scales. *The Quarterly Journal of Mathematics*. Vol. 76(1), pp. 251-263, 2025. <https://doi.org/10.1093/qmath/haae069>)); and Chou, Haag, Huryn, and Ledoan relate the error term in the Hardy-Littlewood prime-pair formula to an exponential sum over primes ((L. Chou, S. Haag, J. Huryn, A. Ledoan. The error term in counting prime pairs. *Journal of Number Theory*. Vol. 278, pp. 422-450, 2026. <https://doi.org/10.1016/j.jnt.2025.04.009>)).

The missing step for the selector model in this paper is not the local factor itself but a theorem controlling dependence among the different candidate complements $N-r$. The singular series corrects congruence biases; it does not by itself justify independence. This is why the Poisson calculation

below is used only as a qualitative heuristic and is tested numerically rather than promoted to a theorem.

Position of the present work

Languasco and Zaccagnini study the number of Goldbach representations ((A. Languasco, A. Zaccagnini. The number of Goldbach representations of an integer. *Proceedings of the American Mathematical Society*. Vol. 140(3), pp. 795-804, 2012. <https://doi.org/10.1090/S0002-9939-2011-10957-2>)). Bauer and Wang treat binary Goldbach with primes restricted to arithmetic progressions with large modulus ((C. Bauer, Y. Wang. The binary Goldbach conjecture with primes in arithmetic progressions with large modulus. *Acta Arithmetica*. Vol. 159(3), pp. 227-243, 2013. <https://doi.org/10.4064/aa159-3-2>)), while Bauer applies a sparse-moduli large-sieve inequality to the Goldbach problem ((C. Bauer. Large sieve inequality with sparse sets of moduli applied to Goldbach conjecture. *Frontiers of Mathematics in China*. Vol. 12(2), pp. 261-280, 2017. <https://doi.org/10.1007/s11464-016-0527-x>)). Oliveira e Silva, Herzog, and Pardi give large-scale computational verification of even Goldbach and prime gaps ((T. Oliveira e Silva, S. Herzog, S. Pardi. Empirical verification of the even Goldbach conjecture and computation of prime gaps up to 4×10^{18} . *Mathematics of Computation*. Vol. 83(288), pp. 2033-2060, 2014. <https://doi.org/10.1090/S0025-5718-2013-02787-1>)). Platt and Trudgian study Linnik's approximation to Goldbach and related questions ((D. J. Platt, T. S. Trudgian. Linnik's approximation to Goldbach's conjecture, and other problems. *Journal of Number Theory*. Vol. 153, pp. 54-62, 2015. <https://doi.org/10.1016/j.jnt.2015.01.008>)). Friedlander, Goldston, Iwaniec, and Suriajaya specifically examine exceptional zeros in the Goldbach problem ((J. B. Friedlander, D. A. Goldston, H. Iwaniec, A. I. Suriajaya. Exceptional zeros and the Goldbach problem. *Journal of Number Theory*. Vol. 233, pp. 78-86, 2022. <https://doi.org/10.1016/j.jnt.2021.06.004>)). Distributional questions for primes in short and logarithmic intervals have also been developed by Montgomery-Soundararajan, Bazzanella-Languasco-Zaccagnini, and Mastrostefano ((H. L. Montgomery, K. Soundararajan. Primes in short intervals. *Communications in Mathematical Physics*. Vol. 252, pp. 589-617, 2004. <https://doi.org/10.1007/s00220-004-1222-4>)). ((D. Bazzanella, A. Languasco, A. Zaccagnini. Prime numbers in logarithmic intervals. *Transactions of the American Mathematical Society*. Vol. 362(5), pp. 2667-2684, 2010. <https://doi.org/10.1090/S0002-9947-09-05009-0>)). ((D. Mastrostefano. Short intervals containing a prescribed number of primes. *Journal of Number Theory*. Vol. 197, pp. 49-61, 2019. <https://doi.org/10.1016/j.jnt.2018.06.013>)), while Languasco studies a conditional short-interval exceptional-set problem ((A. Languasco. A conditional result on the exceptional set for Hardy-Littlewood numbers in short intervals. *International Journal of Number Theory*. Vol. 5(6), pp. 933-951, 2009. <https://doi.org/10.1142/S179304210900247X>)).

Against this background, the research question here is deliberately narrower: how far can a simple two-stage residual construction be pushed from a rigorous prime-in-interval input, and what extra selector information would be required to force the unit transition $2n \rightarrow 2n+2$? The proved contribution is a generated-family lower bound. The Cramér-scale improvement is conditional, and the density-one unit-step statement is only a model prediction.

Known Results and Heuristic Background

Baker-Harman-Pintz short-interval theorem

Baker, Harman, and Pintz prove that there exists x_0 such that every $x \geq x_0$ has a prime in $[x - x^{0.525}, x]$ ((R. C. Baker, G. Harman, J. Pintz. The difference between consecutive primes, II. *Proceedings of the London Mathematical Society*. Vol. 83(3), pp. 532-562, 2001. <https://doi.org/10.1112/plms/83.3.532>)). The strict one-sided statements needed by the construction follow as corollaries. For a prime strictly below y , apply the theorem at $x = y - 1$; after enlarging the constant and threshold, this gives $y - B y^{0.525} < p < y$. For a prime strictly above y , fix $A > 1$ and put

$z=y+A y^{0.525}$. For sufficiently large y , $z^{0.525}<A y^{0.525}$, hence $z-z^{0.525}>y$. Applying Baker-Harman-Pintz at z gives $y<p\leq y+A y^{0.525}$. After enlarging constants and thresholds, one may therefore use a common B for the strict-below and strict-above searches.

Cramér gap conjecture and model

Cramér's 1936 probabilistic model treats integers near x as independently prime with probability about $1/\log x$ and motivates maximal gaps on the $(\log x)^2$ scale ((H. Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*. Vol. 2(1), pp. 23-46, 1936. <https://doi.org/10.4064/aa-2-1-23-46>)). Arithmetic correlations invalidate literal independence, and Granville's analysis explains why local congruence structure changes refined predictions ((A. Granville. Harald Cramér and the distribution of prime numbers. *Scandinavian Actuarial Journal*. Vol. 1995(1), pp. 12-28, 1995. <https://doi.org/10.1080/03461238.1995.10413946>)). In this paper the statement 'a prime occurs within $A(\log y)^2$ on either side of y ' is therefore introduced explicitly as an unproved assumption, not as Cramér's theorem.

Hardy-Littlewood Goldbach heuristic and singular series

For a large even integer N , the Hardy-Littlewood heuristic predicts a Goldbach representation count with leading factor $2C_2 S(N)$, where C_2 is the twin-prime constant and $S(N)$ is the local correction ((G. H. Hardy, J. E. Littlewood. Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes. *Acta Mathematica*. Vol. 44, pp. 1-70, 1923. <https://doi.org/10.1007/BF02403921>)). In the normalization used below,

$$C_2 := \prod_{p>2} \{1 - 1/(p-1)^2\},$$

$$S(N) := \prod_{p|N, p>2} \{p-1\}/(p-2).$$

The factor $S(N)$ is deterministic arithmetic data. A Poisson law for the selector count would require substantially more than inserting $S(N)$ into a first-moment estimate. Conditional results such as Leung's show how Poisson behavior can arise when a suitably uniform Hardy-Littlewood prime-tuple hypothesis is assumed ((S.-K. Leung. Pseudorandomness of primes at large scales. *The Quarterly Journal of Mathematics*. Vol. 76(1), pp. 251-263, 2025. <https://doi.org/10.1093/qmath/haae069>)); no such theorem is assumed here.

Construction Method and Prime Distribution Bounds

Let $2n$ be a sufficiently large even integer. Set $X=2n-5$. Choose a prime p' in the first-stage interval and define the compressed residual

$$M := 2n - p'.$$

A second prime q' is selected above M . Since M and q' are odd, $q'-M$ is even, so there is a positive integer c with

$$q' = M + 2c,$$

and consequently

$$p' + q' = p' + (2n - p') + 2c = 2n + 2c.$$

This is the substantive two-stage construction.

Relation to the original shift parametrization

If an input Goldbach representation $p+q=2n$ is available, the same first-stage prime can be written $p'=p+2k$ with $k=(p'-p)/2 \in \mathbb{Z}$. Since $p=2n-q$,

$$q - 2k = q - (p'-p) = p+q-p' = 2n-p' = M.$$

Hence $q'=q-2k+2c$. The identities in p,q,k are therefore an algebraic parametrization of the direct construction. After p' is selected, the residual M and the second prime do not depend on the particular starting representation.

Unconditional short-interval bound

By the Baker-Harman-Pintz theorem and the strict one-sided corollaries stated above, there are constants $B>0$ and y_0 such that for every $y\geq y_0$ the required prime searches can be completed within $B y^\theta$, with $\theta=0.525$ ((R. C. Baker, G. Harman, J. Pintz. The difference between consecutive primes, II. Proceedings of the London Mathematical Society. Vol. 83(3), pp. 532-562, 2001. <https://doi.org/10.1112/plms/83.3.532>)). For sufficiently large n choose

$$p' \in (X - B X^\theta, X), \quad X=2n-5.$$

Then

$$5 < M = 2n-p' < 5 + B(2n-5)^\theta.$$

Let M_0 be a threshold beyond which the upper-side corollary is valid. If $M\geq M_0$, choose a prime q' with $M < q' \leq M+B M^\theta$. Writing $q'=M+2c$ gives

$$0 < c \leq (B/2)M^\theta \leq (B/2)[5+B(2n-5)^\theta]^\theta.$$

If $5 < M < M_0$, only finitely many odd residuals occur. Let $P^+(M)$ be the least prime greater than M and define

$$C_0 := \max_{\{5 < M < M_0, M \text{ odd}\}} (P^+(M)-M)/2.$$

Then $c \leq C_0$ in the bounded range. Since the displayed power scale tends to infinity, C_0 is absorbed for sufficiently large n , yielding

$$c = O(n^{(\theta^2)}).$$

With $\theta=0.525$, $\theta^2=0.275625$.

Worked Numerical Example of the Construction

The following example illustrates the mechanics of the construction and the difference between an arbitrary admissible first-stage choice and a unit-step selector. It uses the normalized heuristic window $G(x)=(\log x)^2$ only for illustration; it is not a Baker-Harman-Pintz certificate. Take $2n=200000$. Then $X=199995$ and $G(X)\approx 148.99$. The prime $p'=199967$ lies in $(X-G(X), X)$. It gives $M=200000-199967=33$. The least prime greater than 33 is $q'=37$, so $q'=M+4=M+2c$ with $c=2$, and

$$199967 + 37 = 200004 = 200000 + 2(2).$$

A different admissible first-stage prime, $p'=199961$, gives $M=39$. The least prime above 39 is $41=M+2$, so $c=1$ and

$$199961 + 41 = 200002.$$

Thus the interval construction itself guarantees a later Goldbach even, while the selector question asks whether one can choose p' so that the later even is the immediate successor $2n+2$.

Fundamental Counting and Generated-Family Bound

Starting from any fixed sufficiently large even input, iterate the two-stage construction and denote the successive outputs by $2n_1 < 2n_2 < \dots$. Each output is Goldbach by construction, and the next application depends only on the new even input. Let $a(x)$ count generated values not exceeding x .

From $c=O(n^{(\theta^2)})$, there is a constant C such that while $2n_j \leq x$,

$$2(n_{j+1} - n_j) = 2c_j \leq C x^{(\theta^2)}.$$

A deterministic covering count therefore gives

$$a(x) \gg x^{(1-\theta^2)}.$$

Equivalently, with $d_{\text{gen}}(x) := a(x)/(x/2)$, one has $d_{\text{gen}}(x) \gg x^{(-\theta^2)}$. For $\theta=0.525$,

$$a(x) = \Omega(x^{0.724375}).$$

This lower bound concerns only the family generated by the construction. It is not a density-one theorem for Goldbach numbers and is not a proof of binary Goldbach.

Corrected Two-Scale Cramér-Type Analysis

The fixed-exponent argument evaluates the second prime search at the compressed residual M . The same scale separation is essential under a Cramér-type hypothesis. Motivated by Cramér's model and later refinements ((H. Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*. Vol. 2(1), pp. 23-46, 1936. <https://doi.org/10.4064/aa-2-1-23-46>)) ((A. Granville. Harald Cramér and the distribution of prime numbers. *Scandinavian Actuarial Journal*. Vol. 1995(1), pp. 12-28, 1995. <https://doi.org/10.1080/03461238.1995.10413946>)) ((A. Granville, A. Lumley. Primes in short intervals: Heuristics and calculations. *Experimental Mathematics*. Vol. 32(2), pp. 378-404, 2023. <https://doi.org/10.1080/10586458.2021.1927256>)), assume explicitly that there exist $A > 0$ and Y_0 such that every $y \geq Y_0$ has a prime in $(y - A(\log y)^2, y)$ and a prime in $(y, y + A(\log y)^2]$.

Choose p' in $(X - A(\log X)^2, X)$. Then

$$5 < M < 5 + A(\log(2n-5))^2,$$

so $M = O((\log n)^2)$. If $M \geq Y_0$, choose q' with $M < q' \leq M + A(\log M)^2$; then

$$c \leq (A/2)(\log M)^2 = O((\log \log n)^2).$$

If $5 < M < Y_0$, the finite set of possible residuals is bounded by a constant C_1 exactly as in the unconditional argument. Hence

$$c = O((\log \log n)^2).$$

The same covering count yields the conditional bounds

$$a(x) = \Omega(x/(\log \log x)^2), \quad d_{\text{gen}}(x) \gg 1/(\log \log x)^2.$$

These statements are conditional generated-family bounds; they do not imply full Goldbach or generated density one.

Limits of the Nested Max-Gap Certificate for Unit Steps

Let $G(x)$ be a nondecreasing gap function that certifies the two prime searches at the relevant scales. The first stage gives $5 < M < 5 + G(2n-5)$, and the second gives

$$2c \leq G(M) \leq G(5 + G(2n-5)).$$

Therefore

$$G(5 + G(2n-5)) < 4$$

is sufficient for this particular worst-case bound to force $c=1$, because $2c$ is a positive even integer. If G is nondecreasing and unbounded, the displayed sufficient condition eventually fails. This only shows that this nested worst-case certificate stops forcing $c=1$; it does not rule out a selector argument using the multiplicity of first-stage primes.

A Heuristic Prime-Selector Model for the Unit-Step Transition

Selector count

Let $G(x)$ be the first-stage window length and put $N=2n+2$. A unit step requires $q'=M+2$ and $p'+q'=N$. Writing $q'=r$, both r and $N-r$ must be prime, and the first-stage window implies $7 < r \leq 7 + G(2n-5)$. Define

$$S_G(n) := \sum_{\{7 < r \leq 7 + G(2n-5), r \text{ prime}\}} 1_{\{N-r \text{ is prime}\}}.$$

Then $S_G(n) > 0$ is exactly the event that the allowed window contains a unit-step selector.

First-moment heuristic and local correction

A window of length G contains heuristically about $G/\log G$ prime candidates. For each candidate r , the complementary integer $N-r$ has size about N . Using the Hardy-Littlewood local factor ((G. H. Hardy, J. E. Littlewood. Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes. Acta Mathematica. Vol. 44, pp. 1-70, 1923. <https://doi.org/10.1007/BF02403921>)), the heuristic mean is

$$\lambda(n) := 2 C_2 S(N) G(2n-5) / [\log N \cdot \log G(2n-5)].$$

The singular series accounts for congruence obstructions. It does not establish independence among the events $\{N-r \text{ is prime}\}$; modern singular-series and conditional Poisson results likewise distinguish local corrections from a full independence theorem ((D. A. Goldston, A. I. Suriajaya. The error term in the Cesàro mean of the prime pair singular series. Journal of Number Theory. Vol. 227, pp. 144-157, 2021. <https://doi.org/10.1016/j.jnt.2021.03.004>)) ((S.-K. Leung. Pseudorandomness of primes at large scales. The Quarterly Journal of Mathematics. Vol. 76(1), pp. 251-263, 2025. <https://doi.org/10.1093/qmath/haae069>)) ((L. Chou, S. Haag, J. Huryn, A. Ledoan. The error term in counting prime pairs. Journal of Number Theory. Vol. 278, pp. 422-450, 2026. <https://doi.org/10.1016/j.jnt.2025.04.009>)). Consequently the approximation $P(S_G(n)=0) \approx e^{-\lambda(n)}$ is used only as a model.

For the normalized heuristic $G(x)=(\log x)^2$, $\lambda(n)$ is of order $C_2 S(2n+2) \log(2n)/\log \log(2n)$, so the model predicts a decreasing zero probability. This is not a proved density-one statement.

Corrected large-n singular-series illustration

At an odd prime p , averaging the factor $(p-1)/(p-2)$ over even N gives $1+1/[p(p-2)]$. Hence the product of mean local factors is $1/C_2 \approx 1.5147801281$, rather than 1. For $n=10^{69}$, direct factorization gives $S(2n+2)=1.4983622183666705$. With natural logarithms and $G(2n-5)=(\log(2n-5))^2$:

Quantity	Value
$\log(2n)$	159.5715185971491
$\log \log(2n)$	5.072492214185656
$S(2n+2)$	1.4983622183666705
$G(2n-5)$	25463.0695474003
$\lambda(n)$	31.11724960392055
$e^{-\lambda}$	$3.061612377665808 \times 10^{-14}$

The numerical value is illustrative only because the independence approximation is unproved.

Numerical Experiment

Method

The finite experiment evaluates every integer n from 100,000 through 1,000,000 inclusive, for 900,001 inputs, with $G(x)=(\log x)^2$ and natural logarithms. A sieve of Eratosthenes is constructed through 2,000,100, which covers every primality query in the experiment. For each n , set $N=2n+2$ and enumerate every prime r with $7 < r \leq 7+G(2n-5)$; $S_G(n)$ is the number for which $N-r$ is also prime. The singular factor $S(N)$, the model mean $\lambda(n)$, and $e^{-\lambda(n)}$ are computed from the formulas above.

For the minimal shift, enumerate every odd residual M satisfying $5 < M < 5+G(2n-5)$ for which $p'=2n-M$ is prime. Let $P+(M)$ be the least prime greater than M and define $c(M)=(P+(M)-M)/2$. The recorded quantity is $c_min(n)=\min_M c(M)$. By construction, $c_min(n)=1$ if and only if $S_G(n)>0$, providing an internal consistency check between the two computations.

Across the full range, the empirical mean of $S_G(n)$ is 5.8146990948 and the empirical variance is 8.5190218486. There are 2,985 zero-selector inputs, a frequency of 0.3316663%. The average model mean is 5.2489892324, while the average independent-Poisson zero probability is 1.51714135%. Thus

the model has the correct rough scale for the mean but substantially overpredicts zero selectors on this finite range.

Full distribution of $S_G(n)$

$S_G(n)$	Count	Percent
0	2,985	0.331666%
1	20,091	2.232331%
2	61,813	6.868103%
3	113,914	12.657097%
4	144,838	16.093093%
5	137,101	15.233428%
6	109,656	12.183986%
7	82,507	9.167434%
8	65,170	7.241103%
9	52,095	5.788327%
10	39,837	4.426328%
11	28,148	3.127552%
12	18,253	2.028109%
13	11,224	1.247110%
14	6,248	0.694221%
15	3,263	0.362555%
16	1,580	0.175555%
17	757	0.084111%
18	315	0.035000%
19	125	0.013889%
20	53	0.005889%
21	19	0.002111%
22	5	0.000556%
23	3	0.000333%
24	1	0.000111%

The distribution is concentrated around 4–7 selectors but has a visible right tail through 24 in this range. Its variance exceeds its mean, so a single Poisson law with one fixed parameter would not describe the mixture across n even before candidate dependence is considered.

Distribution of the minimal shift c

c_min	Count	Percent
1	897,016	99.668334%
2	2,976	0.330666%
3	9	0.001000%

The value $c_min=1$ occurs for 897,016 of the 900,001 inputs, or 99.668334%. The remaining 2,985 inputs are exactly the $S_G(n)=0$ cases: 2,976 have $c_min=2$ and 9 have $c_min=3$. No tested input requires $c_min \geq 4$. This strong finite-range concentration is empirical evidence about the chosen window; it is not an asymptotic theorem.

Comparison with Known Exceptional-Set Results

Let $E(x)$ be the number of even $m \leq x$ that are not sums of two primes. Montgomery and Vaughan's power-saving estimate already implies that Goldbach-representable even integers have density one ((H. L. Montgomery, R. C. Vaughan. The exceptional set in Goldbach's problem. *Acta Arithmetica*. Vol. 27, pp. 353-370, 1975.)). Li and Lu obtained later exceptional-set refinements ((H. Li. The exceptional set of Goldbach numbers (II). *Acta Arithmetica*. Vol. 92(1), pp. 71-88, 2000. <https://doi.org/10.4064/aa-92-1-71-88>)) (W. C. Lu. Exceptional set of Goldbach number. *Journal of Number Theory*. Vol. 130(10), pp. 2359-2392, 2010. <https://doi.org/10.1016/j.jnt.2010.03.017>)); Matomäki studied exceptional sets in short intervals ((K. Matomäki. On the exceptional set in Goldbach's problem in short intervals. *Monatshefte für Mathematik*. Vol. 155(2), pp. 167-189, 2008. <https://doi.org/10.1007/s00605-008-0561-1>)); and Grimmelt studied Goldbach numbers in short intervals ((L. Grimmelt. Goldbach numbers in short intervals. *Annali della Scuola Normale Superiore*

di Pisa - Classe di Scienze. Vol. 23(3), pp. 1395-1416, 2022. [https://doi.org/10.2422/2036-2145.202002_001.\)\)](https://doi.org/10.2422/2036-2145.202002_001.))). The 2026 treatment of Bhowmik and Grimmelt gives a current overview together with new major-arc information ((G. Bhowmik, L. Grimmelt. The exceptional set of the Goldbach problem. Analysis Mathematica, 2026. [https://doi.org/10.1007/s10476-026-00191-z.\)\)](https://doi.org/10.1007/s10476-026-00191-z.)))).

The theorem proved here is of a different type. The unconditional bound $\Omega(x^{0.724375})$ counts only values reached by one deterministic construction. It neither improves $E(x)$ nor approaches the strongest density statements about all Goldbach numbers. Its purpose is structural: the two-stage argument makes explicit what follows from a prime-in-interval input alone and isolates the additional selector property needed for unit steps.

Limitations

The paper has five principal limitations. First, the generated-family lower bounds concern only outputs of the construction and do not prove binary Goldbach or generated density one. Second, Baker-Harman-Pintz is asymptotic, so finite residuals must be handled separately and no explicit numerical threshold is claimed here. Third, the $O((\log x)^2)$ prime-gap input is an unproved Cramér-type assumption rather than a theorem. Fourth, the selector calculation uses a first-moment Hardy-Littlewood correction together with a Poisson independence approximation that is not justified for the candidate events in this problem. Fifth, the computation covers only $100,000 \leq n \leq 1,000,000$ with the specific normalization $G(x) = (\log x)^2$; the observed 99.668334% unit-selector frequency and the empirical distributions do not establish any limiting density.

Open Problems and Future Directions

Residual prime-selector conjecture

The selector question can be stated directly. For a chosen window G , ask whether every sufficiently large n admits a prime

$$p' \in (2n-5-G(2n-5), 2n-5)$$

such that $2n-p'+2$ is prime. This remains an unproved conjectural condition. The Poisson model and the finite experiment do not prove it.

Correlation-sensitive analysis

A natural next step is a second-moment or sieve analysis of $S_G(n)$ that tracks dependence among the complements $N-r$. The correlation and small-gap methods of Goldston-Yıldırım and Goldston-Pintz-Yıldırım provide relevant precedent ((D. A. Goldston, C. Y. Yıldırım. Higher correlations of divisor sums related to primes III: small gaps between primes. Proceedings of the London Mathematical Society. Vol. 95(3), pp. 653-686, 2007. [https://doi.org/10.1112/plms/pdm021.\)\)](https://doi.org/10.1112/plms/pdm021.))); ((D. A. Goldston, J. Pintz, C. Y. Yıldırım. Primes in tuples I. Annals of Mathematics. Vol. 170(2), pp. 819-862, 2009. [https://doi.org/10.4007/annals.2009.170.819.\)\)](https://doi.org/10.4007/annals.2009.170.819.))); ((D. A. Goldston, J. Pintz, C. Y. Yıldırım. Primes in tuples II. Acta Mathematica. Vol. 204(1), pp. 1-47, 2010. [https://doi.org/10.1007/s11511-010-0044-9.\)\)](https://doi.org/10.1007/s11511-010-0044-9.)))), while the later bounded-gap literature supplies complementary sieve techniques ((Y. Zhang. Bounded gaps between primes. Annals of Mathematics. Vol. 179(3), pp. 1121-1174, 2014. [https://doi.org/10.4007/annals.2014.179.3.7.\)\)](https://doi.org/10.4007/annals.2014.179.3.7.))); ((D. H. J. Polymath. Variants of the Selberg sieve, and bounded intervals containing many primes. Research in the Mathematical Sciences. Vol. 1, Article 12, 2014. [https://doi.org/10.1186/s40687-014-0012-7.\)\)](https://doi.org/10.1186/s40687-014-0012-7.))); ((J. Maynard. Small gaps between primes. Annals of Mathematics. Vol. 181(1), pp. 383-413, 2015. [https://doi.org/10.4007/annals.2015.181.1.7.\)\)](https://doi.org/10.4007/annals.2015.181.1.7.)))).

Minimal-shift statistics

The computed distribution of $c_{\min}$ suggests studying its moments, tails, dependence on $S(2n+2)$, and sensitivity to the first-stage window. Such questions are numerical or conjectural at present. Large-

scale Goldbach and prime-gap computation provides a useful benchmark for designing those experiments ((T. Oliveira e Silva, S. Herzog, S. Pardi. Empirical verification of the even Goldbach conjecture and computation of prime gaps up to 4×10^{18} . *Mathematics of Computation*. Vol. 83(288), pp. 2033-2060, 2014. <https://doi.org/10.1090/S0025-5718-2013-02787-1>)).

Conclusion

The construction is best viewed as a direct two-stage residual procedure. From a sufficiently large even input $2n$, it selects p' near $2n-5$, forms $M=2n-p'$, and selects the next controlled prime above M , thereby producing a later Goldbach even $2n+2c$. The p,q,k notation is an equivalent parametrization when an initial Goldbach representation is available, but it is not the mechanism of the construction.

Using the Baker-Harman-Pintz exponent $\theta=0.525$, with the strict one-sided searches derived as corollaries and the bounded residual range treated separately, the construction gives $c=O(n^{0.275625})$ and $\Omega(x^{0.724375})$ generated values. Under an explicitly unproved Cramér-type gap hypothesis it gives $c=O((\log \log n)^2)$ and $\Omega(x/(\log \log x)^2)$. The nested max-gap inequality is only a sufficient unit-step certificate.

The selector remains heuristic. The full computation on 900,001 inputs finds $S_G(n)=0$ only 2,985 times and $c_{\min}=1$ in 99.668334% of cases, but the empirical zero rate is materially below the independent-Poisson prediction. That discrepancy reinforces the central unresolved point: local singular-series corrections are not a substitute for a correlation theorem. The proved contribution is therefore the generated-family analysis and the separation of worst-case interval control from the stronger selector behavior required for $c=1$.

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